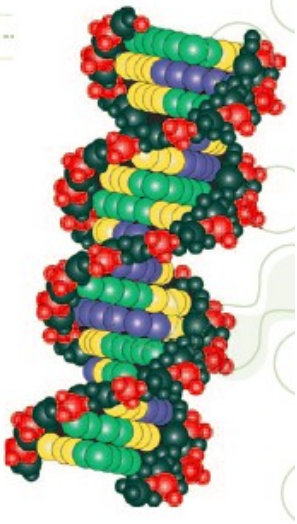
2.1 Molecule mastermind!



Life on Earth is extraordinary for its diversity. However, living things from the smallest bacterium to the largest tree or mammal are all bound by a common thread. Their characteristics are determined by the very same chemical material: a complex molecule called **deoxyribonucleic acid** or **DNA**. The structure of that DNA is similar in every one of these vastly different organisms.

If you want to see what DNA looks like, you could carry out the following activity.

figure 2.1 The double helix of DNA is an amazing molecule; it is the fundamental biological chemical that contains the blueprint for life itself.

DNA on the inside

DNA was first discovered in 1869, but it was not until 1944 that it was shown that genes were composed of this chemical. Then in 1953 the structure of DNA was worked out. DNA is a long threadlike molecule similar in shape to a rope ladder that has been twisted into a spiral—the **double helix**. The sides of the ladder are composed of alternating sugar and phosphate units. The rungs of the ladder are made up of paired chemical units or bases. There are four different **bases**: **adenine**, **guanine**, **cytosine** and **thymine** (usually referred to as A, G, C and T). Adenine only pairs with thymine and guanine only pairs with cytosine. This kind of pairing is called **complementary base pairing**. These four different bases make up the genetic alphabet. They are used to write the blueprint for every characteristic in your body. The difference in the number and arrangement of these bases also accounts for the differences between all species of living things.

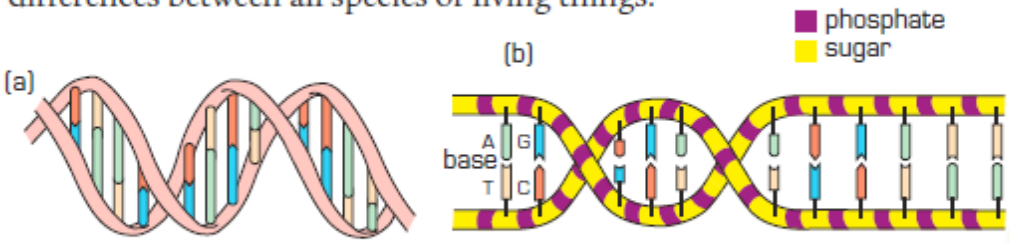


figure 2.2 (a) The DNA molecule showing the double helix structure. (b) DNA 'unwound' to show the relative positions of the bases and the sugar and phosphate molecules.

Genes and chromosomes

Chromosomes are long, thin, threadlike structures made of DNA and are found in the nucleus of cells. They are only really visible during cell division. The cells in the human body each contain 46 chromosomes (in 23 pairs) with the exception of the sperm and egg cells which only contain 23 chromosomes (or one of each pair). Other organisms have different numbers of chromosome pairs in their cells.

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A genome is the complete set of genetic information for a species. In humans this represents about 30 000 genes, made up of about 1 billion base pairs. The Human Genome Project is an international ...dentify every human gene ...ermine the base sequence ... for each of them. The ...nome was completed in ... However, knowing the

Multiplying and dividing

Your life began as a single cell, produced by the fusion of an egg cell and sperm. But now your body is made up of billions of cells. How did they get there? Where did they come from? Every cell in your body originated from this single cell. How?

Cells have the ability to **replicate** (make an exact copy of) the DNA they contain. To do this the bases of the double helix separate from each other, in much the same way as a zip opens, leaving the bases exposed. Within each nucleus there are bases floating

SciFile

If all the chromosomes in a single human cell were unwound and connected end to end they would measure 1 m in length. The 1 m

Questions 2.1

1 Name the chemical that is the genetic blueprint for all living things.

2 Use coloured pencils to draw and label a simple diagram representing the DNA molecule.

3 Explain in your own words what is meant by the term 'complementary base pairing'.

4 In an organism, where is the DNA found?

5 Why is it essential for the chromosomes to replicate before cell division occurs?

6 A gamete or sex cell is haploid. What does this statement mean?

7 **a** A horse has 64 chromosomes. How many chromosomes would be found in the sperm and egg cells of a horse?

b How many chromosomes would be found in the cells of a tomato plant? There are 12 chromosomes in the gametes.

c Which type of cell division produces the sex cells?